CENG 467 – Natural Language Understanding and Generation
Take-Home Midterm Examination

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1 Introduction

1.1 Scope and Objectives

This take-home report studies five natural language understanding and generation problems that cover both analysis and generation settings: sentiment classification, named entity recognition, summarization, machine translation, and language modeling. The common objective across all questions is not only to obtain strong final metrics, but also to compare different representation regimes in a controlled way. Wherever possible, the experiments contrast classical feature-based baselines, supervised neural architectures, and transformer or pretrained contextual models so that the final discussion can separate the effect of model class from the effect of representation quality.

At the time of this draft, all five question sections now have written report content, but they are not equally mature. Questions 1 and 2 remain the most complete direct comparisons, Questions 3 and 5 now have stable capped comparison artifacts and drafted sections, and Question 4 now has a direct transformer-versus-seq2seq comparison with an explicitly noted train-budget mismatch because the transformer is a pretrained reference. This staged approach keeps the written analysis aligned with verified artifacts rather than speculative intermediate results.

1.2 Technical Setup

All experiments are managed from YAML configuration files, use a fixed random seed of 42 for reproducibility, and write timestamped run directories that include the resolved configuration, package environment snapshot, metrics, and analysis artifacts. The implementation stack combines Hugging Face Datasets for dataset access, scikit-learn and related linear modeling tools for classical baselines, PyTorch for custom neural models, and transformers for pretrained contextual encoders. Report assembly follows the same principle of artifact stability: figures copied into `report/figures/` and tables placed in `report/tables/` are sourced from completed experiment outputs rather than ad hoc terminal observations.

Table 1 summarizes the five questions, their datasets, and the representative model families covered by the report. Some sections already contain finished direct comparisons, while others currently document strong capped or baseline-first reference points that can be extended later.

Q	Task	Dataset	Representative Models
1	Text classification	IMDb	TF-IDF+LR/SVM, BiLSTM, DistilBERT
2	Named entity recognition	CoNLL-2003	CRF, BiLSTM-CRF, BERT
3	Summarization	CNN/DailyMail	TextRank, DistilBART
4	Machine translation	Multi30k	Seq2Seq+Attn, Opus-MT
5	Language modeling	WikiText-2	Trigram, LSTM, distilGPT-2

Table 1: Task and dataset overview for the take-home report.

1.3 Report Organization

Section 2 presents Question 1, which analyzes representation learning for IMDb sentiment classification and compares TF-IDF baselines, a word-level BiLSTM, and DistilBERT un-

der a matched larger-budget setup. Section 3 presents Question 2, which studies named entity recognition on CoNLL-2003 with CRF, BiLSTM-CRF, and BERT token classification. Section 4 presents Question 3, which now includes a direct capped TextRank-versus-DistilBART summarization comparison. Section 5 presents Question 4, which now compares a pretrained transformer reference against a custom seq2seq-plus-attention baseline on Multi30k while making the train-budget mismatch explicit. Section 6 presents Question 5, which now compares trigram, LSTM, and GPT-style transformer language-model references on a matched capped WikiText-2 setup. The conclusion section synthesizes the current cross-task findings while remaining provisional where later extensions could still change the strength of the final comparative claims.

2 Question 1: Representation Learning in Text Classification

2.1 Dataset Description

Question 1 uses the IMDb movie-review sentiment benchmark [7], which contains 50,000 balanced reviews split into 25,000 original training examples and 25,000 original test examples. The project pipeline normally applies a stratified 80/20 split to the training portion, but the matched larger-budget comparison used a capped setting to keep the neural runs tractable on the available hardware. Concretely, 4,000 reviews were first sampled from the original training split and then divided into 3,200 training and 800 validation examples, while 2,000 reviews were sampled from the original test split for final evaluation. Because the label distribution is balanced, accuracy and macro-F1 move closely together and both are useful summary metrics.

2.2 Preprocessing Pipeline

All Q1 models share the same base text-normalization pipeline: HTML tags are stripped, text is lowercased, non-alphanumeric characters are removed, whitespace is normalized, and documents are truncated to 256 tokens. TF-IDF and BiLSTM consume these normalized word-level sequences directly, whereas DistilBERT applies WordPiece tokenization after the shared cleaning stage.

The preprocessing sweep was intentionally run as a cheaper validation-only study on TF-IDF + Logistic Regression rather than as a full end-to-end rerun of every model. Table 2 shows that keeping stopwords with lowercasing and 50,000 features (Experiment B) tied with preserving case (Experiment D), both reaching validation macro-F1 0.8178. Since the lowercase configuration is simpler and aligns naturally with the uncased DistilBERT backbone, the final larger-budget experiments kept the lowercase + keep-stopwords setting.

Exp.	Stopwords	Lowercase	Max Features	Val. Acc.	Val. Macro-F1
B	Keep	Yes	50,000	0.8200	0.8178
D	Keep	No	50,000	0.8200	0.8178
C	Remove	Yes	100,000	0.8150	0.8125
A	Remove	Yes	50,000	0.8000	0.7980

Table 2: Validation-only TF-IDF + Logistic Regression preprocessing sweep on a capped 800-train/200-validation subset.

2.3 Models

TF-IDF + Logistic Regression / SVM. The sparse baselines use unigram and bigram TF-IDF features with 50,000 maximum features and regularization strength $C = 1.0$. Two classifiers were tested on top of the same representation: Logistic Regression as a probabilistic linear baseline and a linear SVM as a max-margin variant.

BiLSTM. The word-level neural baseline uses learned 128-dimensional embeddings, a single bidirectional LSTM with hidden size 128, dropout 0.3, batch size 64, and Adam optimization with learning rate 10^{-3}. Early stopping monitors validation macro-F1 with patience 2 and training can run for up to 6 epochs. In the current implementation, this model does not use pretrained GloVe initialization, so its representation quality depends entirely on the supervised IMDb subset.

DistilBERT. The contextual baseline fine-tunes `distilbert-base-uncased` [11] with batch size 16, learning rate 2×10^{-5}, weight decay 0.01, warmup ratio 0.1, and maximum sequence length 256. The completed larger-budget run converged within 2 epochs and produced the strongest validation and test scores among all finished Q1 models.

2.4 Results

Table 3 summarizes the matched 4k/2k comparison. DistilBERT is the strongest finished model with test macro-F1 0.8793 and test accuracy 0.8795. It outperforms TF-IDF + SVM by roughly 0.028 macro-F1 points, TF-IDF + LR by roughly 0.039, and BiLSTM by roughly 0.178. The classical sparse baselines remain competitive, especially the linear SVM, while the randomly initialized BiLSTM trails substantially behind both TF-IDF variants and the transformer model.

Model	Val. Acc.	Val. Macro-F1	Test Acc.	Test Macro-F1
DistilBERT	0.8750	0.8749	0.8795	0.8793
TF-IDF + SVM	0.8638	0.8637	0.8510	0.8510
TF-IDF + LR	0.8650	0.8650	0.8400	0.8400
BiLSTM	0.7413	0.7386	0.7025	0.7011

Table 3: Overall Q1 performance on the matched larger-budget setting with a 4,000-example capped train split, an 800-example validation split, and a 2,000-example test split.

Figure 1 visualizes the same ranking. The gap between DistilBERT and TF-IDF is real but modest, whereas the gap between both of those approaches and BiLSTM is much larger. This suggests that pretrained contextual representations provide the most value on the capped budget, while a lightweight sparse baseline still offers a strong accuracy-to-cost trade-off.

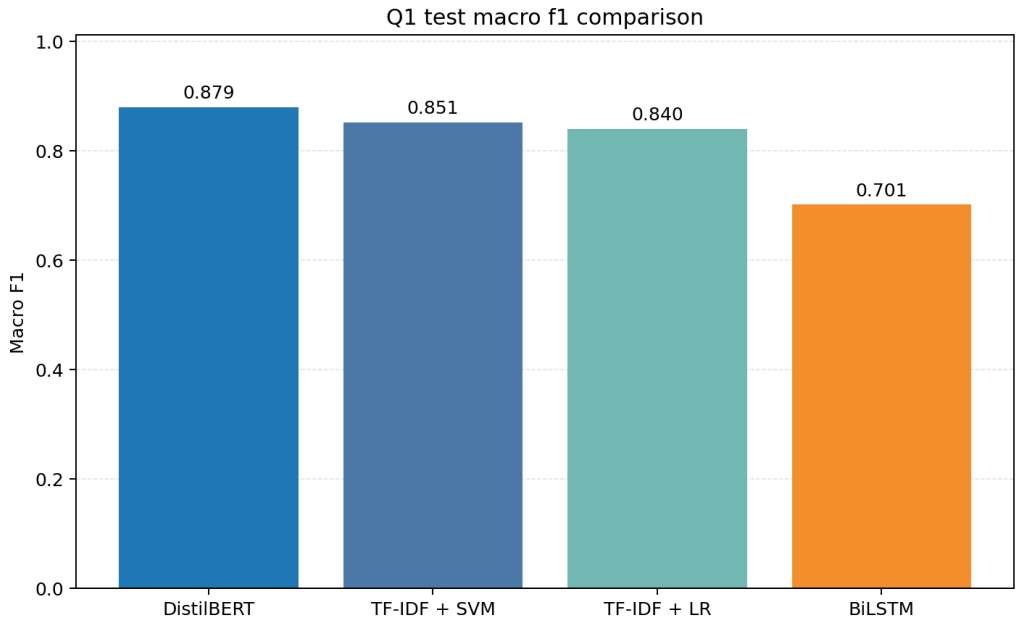

Figure 1: Test macro-F1 comparison for the matched larger-budget Q1 runs.

The current training pipeline does not persist per-epoch histories for the larger-budget neural runs, so the comparison focuses on final validation/test metrics and confusion patterns rather than explicit training curves. Figure 2 shows the test confusion matrix for the best model. DistilBERT remains fairly balanced across both classes, but it recovers positive reviews more strongly than negative ones, with positive-class recall 0.918 versus negative-class recall 0.841.

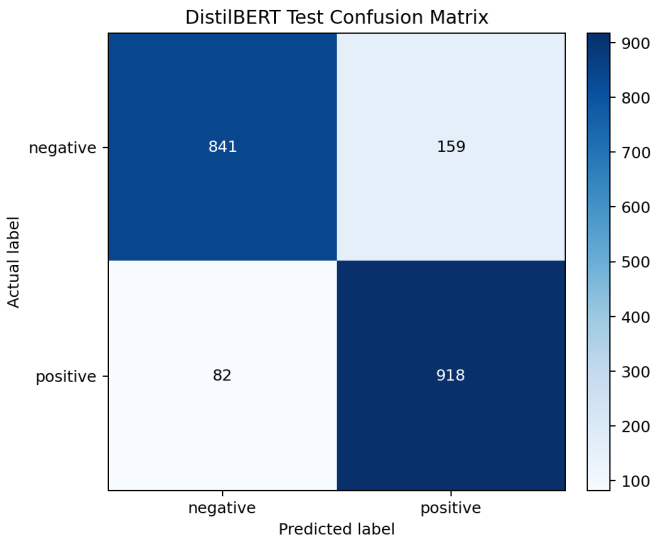

Figure 2: Test confusion matrix for the strongest Q1 model, DistilBERT.

2.5 Error Analysis

The exported misclassification artifacts indicate that negation remains the dominant failure mode across all representation families. For DistilBERT, 4 of the top 5 sampled test errors fall into the `negation_handling` category, usually in long reviews where local positive phrases are later overturned by a negative judgment. The TF-IDF baselines show a broader mix of negation, mixed sentiment, and domain-specific ambiguity, which is consistent with lexical models that cannot fully resolve discourse-level polarity shifts. BiLSTM is the most brittle model: its test error summary is still dominated by negation, but it also struggles with very long reviews and mixed sentiment, and its confusion matrix shows a much stronger drop in positive recall than the other approaches.

This qualitative pattern matches the aggregate metrics. DistilBERT's contextual encoder is better at following sentiment reversals across long documents, TF-IDF remains surprisingly robust when the signal is lexical and local, and the supervised-only BiLSTM does not learn a representation strong enough to close the gap. In practical terms, the main unresolved Q1 errors are no longer simple vocabulary misses but cases where polarity depends on clause interaction, contrast, or delayed sentiment cues.

2.6 Discussion

The final larger-budget ranking for Question 1 is clear: DistilBERT is the strongest model, TF-IDF + SVM is the strongest lightweight baseline, TF-IDF + LR is close behind, and BiLSTM is the weakest of the finished runs. The most important takeaway is that representation choice matters more than small preprocessing differences on this task. The preprocessing sweep changed validation macro-F1 only marginally, whereas switching from sparse TF-IDF features to contextual transformer embeddings produced the largest reliable gain.

At the same time, the results do not support a naive "neural is always better" conclusion. Moving from sparse features to a randomly initialized word-level BiLSTM did not help; in fact, it substantially hurt performance on the matched budget. This makes the comparison more informative: the benefit comes from contextual pretraining rather than from replacing linear classifiers with a neural sequence model by itself. For the final report claim, DistilBERT should be presented as the best-performing Q1 model, TF-IDF + SVM as a strong and efficient baseline, and BiLSTM as evidence that dense representations without pretraining or further tuning are not sufficient on their own.

3 Question 2: Named Entity Recognition

3.1 Dataset Description

Question 2 uses the English CoNLL-2003 named entity recognition benchmark [14] with the IOB2 tagging scheme and four entity types: PER, ORG, LOC, and MISC. The local environment could not load the legacy script-based dataset entry directly, so the experiments used the Hugging Face mirror `tomaarsen/conll2003`, which preserves the same token-level label inventory. The final split sizes were 14,041 training sentences, 3,250 validation sentences, and 3,453 test sentences. Model selection focused on strict entity-level precision, recall, and F1 under the IOB2 scheme, while token accuracy was retained as a supplementary diagnostic metric.

3.2 Preprocessing and BIO Alignment

All models consume the original tokenized sentence boundaries from CoNLL-2003. For the feature-based CRF, each token is represented by local orthographic and contextual cues including the lowercased token form, short prefixes and suffixes, case indicators, digit and hyphen flags, token length, and previous or next token features with beginning-of-sequence and end-of-sequence markers. The BiLSTM-CRF model operates on word-level token ids with padding and masking so that sequence length is preserved during recurrent encoding and CRF decoding.

The BERT model uses `bert-base-cased` because capitalization is informative for named entities. Word-level gold tags are aligned to subword tokenization by assigning the original tag only to the first subword of a word and setting special tokens or continuation subwords to `-100`, which removes them from the loss. This keeps optimization aligned with word-level evaluation while still allowing the model to benefit from subword segmentation.

3.3 Models

Feature CRF. The classical baseline is a linear-chain CRF [5] trained with the `lbfgs` optimizer, $c_1 = 0.1$, $c_2 = 0.1$, 100 maximum iterations, and all transitions enabled. This model relies entirely on the engineered token and context features described above.

BiLSTM-CRF. The hybrid neural baseline uses 100-dimensional token embeddings, a single bidirectional LSTM layer with hidden size 128, dropout 0.3, batch size 32, and AdamW optimization with learning rate 0.001. The model was trained for up to eight epochs with early stopping on validation F1. The CRF output layer preserves global transition constraints during decoding.

BERT Token Classification. The transformer baseline fine-tunes `bert-base-cased` [2] with maximum sequence length 128, batch size 8, learning rate 5×10^{-5}, weight decay 0.01, warmup ratio 0.1, and early stopping patience 2. The strongest full-data run completed after two epochs and clearly outperformed the non-transformer baselines.

3.4 Results

Table 4 summarizes the overall validation and test metrics. BERT produced the strongest result by a wide margin, reaching test F1 0.9062. The feature-based CRF remained a strong non-transformer baseline at test F1 0.7948. BiLSTM-CRF improved over its smoke-test behavior but still lagged behind both alternatives at test F1 0.7145.

Model	Val. P	Val. R	Val. F1	Test P	Test R	Test F1	Test Acc.
BERT	0.9525	0.9509	0.9517	0.9028	0.9095	0.9062	0.9807
Feature CRF	0.8939	0.8596	0.8765	0.8131	0.7773	0.7948	0.9572
BiLSTM-CRF	0.8292	0.7625	0.7945	0.7674	0.6684	0.7145	0.9373

Table 4: Overall Q2 NER performance on the validation and test splits.

Table 5 breaks the test results down by entity type. The support column corresponds to the number of gold entities in the test set. BERT dominates every entity category,

with especially strong performance on PER and LOC. MISC is the hardest category for all models and remains the main source of residual error even in the strongest system.

Model	Entity	Precision	Recall	F1	Support
BERT	LOC	0.9269	0.9269	0.9269	1668
Feature CRF	LOC	0.8442	0.8249	0.8344	1668
BiLSTM-CRF	LOC	0.7991	0.7680	0.7832	1668
BERT	MISC	0.7693	0.8077	0.7880	702
Feature CRF	MISC	0.7756	0.7436	0.7593	702
BiLSTM-CRF	MISC	0.7157	0.6239	0.6667	702
BERT	ORG	0.8873	0.8958	0.8916	1661
Feature CRF	ORG	0.7711	0.6773	0.7212	1661
BiLSTM-CRF	ORG	0.7359	0.6273	0.6773	1661
BERT	PER	0.9552	0.9499	0.9526	1617
Feature CRF	PER	0.8351	0.8454	0.8402	1617
BiLSTM-CRF	PER	0.7873	0.6271	0.6981	1617

Table 5: Per-entity test metrics for the completed Q2 runs.

Figure 3 visualizes the per-entity test F1 comparison. The same pattern is visible across all entity types: contextualized BERT representations yield the best scores, CRF remains competitive without pretrained embeddings, and the recurrent hybrid model underperforms despite having a structured decoder.

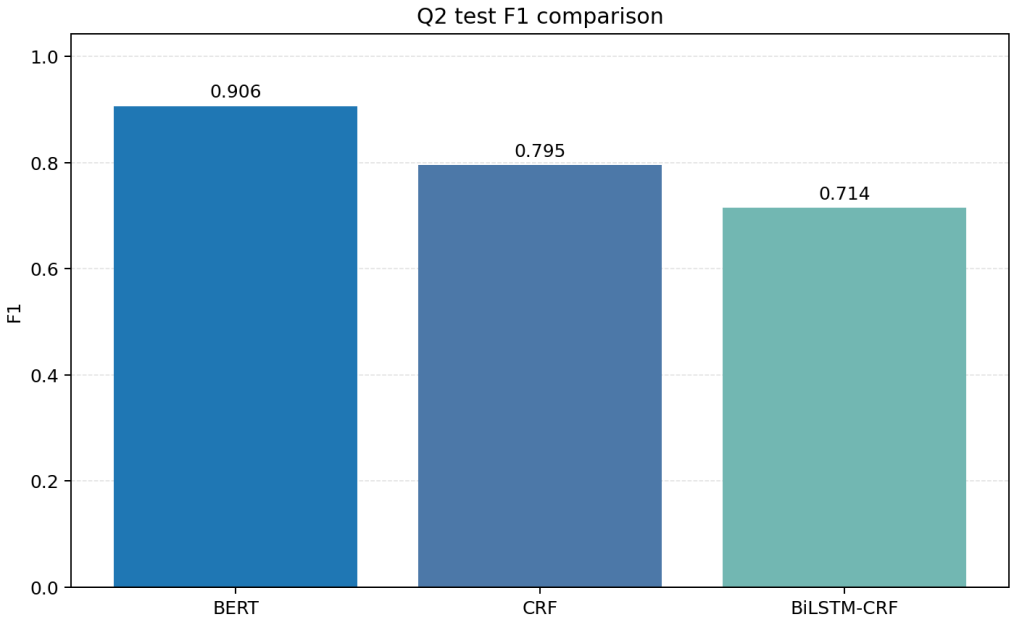

Figure 3: Per-entity test F1 comparison for the completed Q2 CRF, BiLSTM-CRF, and BERT runs.

3.5 Error Analysis

The exported error-analysis artifacts show qualitatively different failure modes across the three models. For the feature CRF, the most common test confusion is `B-ORG` → `O`,

which indicates missed organization boundaries and weaker recall on organization names. BiLSTM-CRF fails most often on person entities, with `B-PER` → `O` as its dominant confusion; this is consistent with its lower recall and suggests that the recurrent encoder did not exploit enough context to recover entity starts reliably. BERT still makes mistakes, but its main confusion pattern is `O` → `I-MISC`; in practice this means that the model occasionally overextends rare miscellaneous entities while still preserving much stronger overall entity recall and boundary quality than the alternatives.

The per-entity table supports this interpretation. Under BERT, MISC remains the weakest class at F1 0.7880, whereas PER and LOC both exceed 0.92 F1. This suggests that contextual embeddings largely solve boundary and lexical ambiguity for common entity categories, while rarer and semantically broader labels such as MISC still require additional disambiguation.

3.6 Discussion

The final ranking for Question 2 is clear: BERT is the strongest model, the feature-based CRF is the strongest non-transformer baseline, and BiLSTM-CRF is the weakest of the completed full-data runs. This outcome highlights two practical points. First, pretrained contextual representations are substantially more valuable than either engineered local features or a randomly initialized recurrent encoder on this task. Second, a well-configured classical CRF can still outperform a neural hybrid model when the latter is not tuned enough to convert representation power into better sequence decisions.

From a report perspective, the BERT run should anchor the main Q2 claim, while the CRF result should be presented as a surprisingly strong and interpretable baseline. BiLSTM-CRF is still useful as a comparison point because it demonstrates that adding recurrence and a CRF layer alone is not sufficient to close the gap to transformer-based contextualization. If additional Q2 effort becomes available later, the most justified follow-up is a focused BiLSTM-CRF tuning slice or a deeper MISC-specific error analysis rather than another baseline implementation.

4 Question 3: Text Summarization

4.1 Dataset Description

Question 3 compares extractive and abstractive summarization on CNN/DailyMail [12], a news summarization benchmark pairing long-form news articles with short highlight summaries written by editors. The architecture notes propose a 10,000/1,000/1,000 subset for efficient experimentation, but the currently stable head-to-head comparison is a capped run with 20 validation and 20 test examples so that both models can be evaluated on the same articles under the same export contract. The resulting numbers should therefore be read as a matched pilot comparison rather than a final large-budget benchmark.

Even with that limitation, CNN/DailyMail remains an appropriate benchmark for this comparison because the articles are substantially longer than the references and require both content selection and compression. This makes the task a natural contrast between a sentence-ranking baseline such as TextRank and a pretrained abstractive model that can rewrite source content into shorter, reference-style summaries.

4.2 Preprocessing and Method

The current Q3 pipeline performs shared article cleanup, truncates long inputs to 900 words, and evaluates all models with the same lexical metrics. The extractive baseline is a TF-IDF based TextRank summarizer [9]: sentences are treated as graph nodes, pairwise cosine similarities define weighted edges, and PageRank scores determine which three sentences are selected for the final summary. The configuration uses damping 0.85 and allows up to 40 candidate sentences per article.

The abstractive baseline uses a pretrained DistilBART CNN checkpoint from the DistilBART family. It is derived from BART-style sequence-to-sequence pretraining [6] and is used here without additional fine-tuning. Inputs are tokenized up to 512 tokens, and decoding uses beam search with output lengths between 20 and 80 tokens. This design keeps the comparison practical on the current environment while still testing the main assignment question: whether an abstractive model can produce more reference-like summaries than a purely extractive sentence selector.

4.3 Results

Table 6 summarizes the matched two-model comparison. On the test split, DistilBART reaches ROUGE-1 0.4218, ROUGE-2 0.1948, and ROUGE-L 0.2833. Its BLEU and METEOR scores are 0.1355 and 0.4254, while TextRank reaches 0.3095, 0.1042, 0.2045, 0.0674, and 0.3041 respectively. The abstractive model outperforms TextRank on every exported metric on both validation and test.

Model	Split	ROUGE-1	ROUGE-2	ROUGE-L	BLEU	METEOR
TextRank	Validation	0.3378	0.1344	0.2267	0.0962	0.3227
TextRank	Test	0.3095	0.1042	0.2045	0.0674	0.3041
DistilBART	Validation	**0.4221**	**0.1978**	**0.2967**	**0.1782**	**0.3621**
DistilBART	Test	**0.4218**	**0.1948**	**0.2833**	**0.1355**	**0.4254**

Table 6: Matched Q3 comparison on the capped 20-validation/20-test TextRank-versus-DistilBART run. DistilBART outperforms TextRank on every exported metric on both splits.

Figure 4 visualizes the same test-set ranking. DistilBART leads TextRank on every exported metric, with especially clear gaps on METEOR, ROUGE-1, and ROUGE-2.

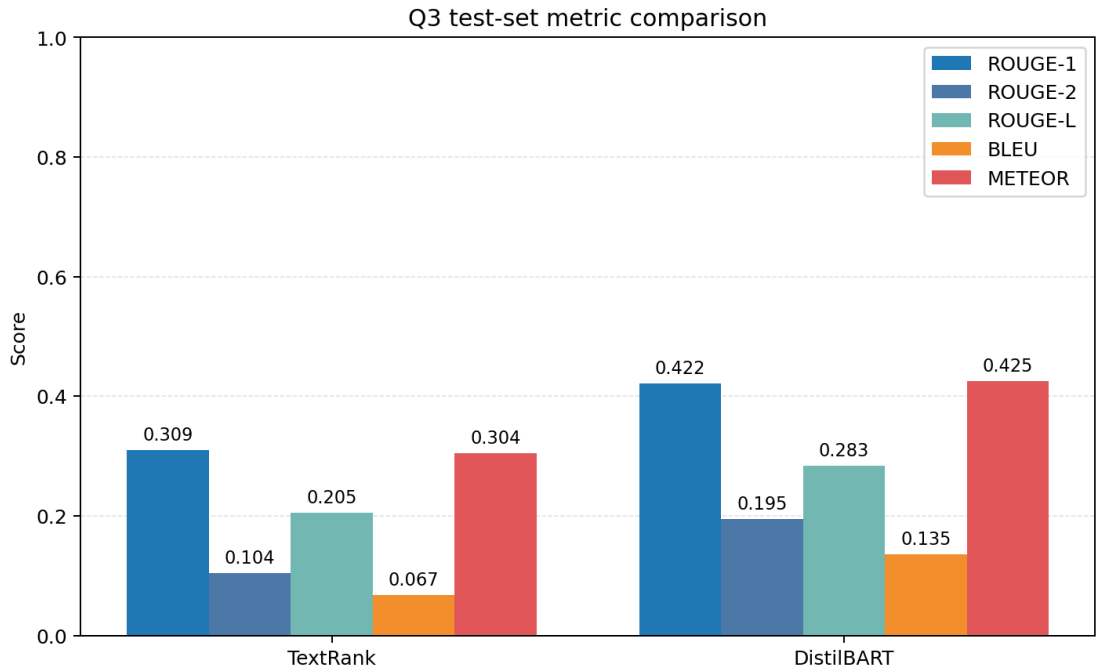

Figure 4: Q3 test-set metric comparison for TextRank and DistilBART.

The largest absolute test improvements are in METEOR (+0.1213), ROUGE-1 (+0.1124), and ROUGE-2 (+0.0906). This pattern suggests that the abstractive model improves both content selection and phrase-level overlap, not only unigram recall. Because the split is still small, these numbers should not be over-generalized. Even so, the results remain consistent across both splits and are materially stronger than the earlier baseline-only Q3 section.

4.4 Qualitative Analysis

The qualitative examples mirror the aggregate scores. In the teen-driver distraction article from the validation split, DistilBART produces a compact summary that preserves the central finding that distraction contributes to nearly 60% of serious teen crashes and correctly identifies passengers and cellphones as the main sources. The corresponding TextRank output is longer, more repetitive, and much less reference-like. A second validation example on Aaron Cook's Moldovan citizenship decision shows the same pattern: both models remain on topic, but the abstractive summary keeps the citizenship, Rio 2016, and Team GB exclusion details while dropping more of the surrounding narrative.

The test split also shows that the abstractive model is not uniformly strong. In the Syrian refugee first-person article, DistilBART stays broadly on topic but still misses key reference details such as the speaker's identity and the specific 12-day journey framing. That limitation matters because it shows that better lexical overlap does not automatically guarantee perfect factual focus. Even so, across the inspected examples DistilBART is generally more concise and closer to the structure of the human-written highlights, whereas TextRank often copies lead sentences or redundant clauses directly from the article.

4.5 Discussion

The current Q3 evidence now supports a clear preliminary conclusion: on this task, a pretrained abstractive baseline is stronger than the extractive TextRank baseline under the same capped evaluation protocol. TextRank remains useful as a cheap and interpretable baseline, and it can preserve source facts when important content is already concentrated in the lead sentences. However, it is weaker at compression and at matching the short editor-style summaries used as references in CNN/DailyMail.

This is still not the final word on Question 3 because the comparison is based on a 20/20 capped split and zero-shot pretrained inference rather than a larger matched benchmark or task-specific fine-tuning. Even so, the section is no longer extractive-only: there is now a stable direct comparison, and the current best Q3 baseline is the pretrained DistilBART model rather than TextRank.

5 Question 4: Machine Translation

5.1 Dataset Description

Question 4 uses the English-to-German Multi30k translation benchmark [3]. In this project, the dataset is accessed through its Hugging Face mirror and treated as a short-caption translation task. That makes it a useful benchmark for testing whether a model can produce fluent target-language sentences while preserving compact visual descriptions from the source caption. The current Q4 evidence now includes two finished systems evaluated on the same capped 100-example validation and 100-example test splits: a pretrained transformer reference and a custom seq2seq-plus-attention baseline.

The comparison is direct at evaluation time but not perfectly budget-matched at training time. The transformer run keeps a 100-example train slice only for artifact-contract consistency and does not perform task-specific fine-tuning, whereas the seq2seq model is trained from scratch on 2000 capped training examples. This distinction matters for interpretation, so the section should be read as a practical pretrained-versus-classical comparison rather than a matched from-scratch study.

5.2 Preprocessing and Model

The stronger system uses a pretrained Opus-MT English-to-German checkpoint from Helsinki-NLP [13]. It is built on a transformer-style translation stack [15]. Source and target casing are preserved, inputs and decoded outputs are capped at 96 tokens, and generation uses beam search with 4 beams and length penalty 1.0. Instead of fine-tuning on Multi30k, this pipeline evaluates the checkpoint directly as a strong practical reference.

The classical comparator is a compact GRU encoder-decoder with additive attention [1]. It uses 128-dimensional token embeddings and hidden states, a single recurrent layer, teacher forcing during training, and a maximum decoded length of 32 tokens. Unlike the transformer reference, this seq2seq model is trained from scratch on a lowercased 2000-example training split. It must therefore learn translation behavior directly from the capped Multi30k data.

Evaluation uses BLEU and ChrF on both validation and test. BLEU gives a familiar n-gram precision view of translation quality, while ChrF is more tolerant of inflectional

and morphological variation and is often informative for German output. Together, the two metrics provide a lightweight but credible baseline suite for the current stage of Q4.

5.3 Results

Table 7 summarizes the finished comparison. The pretrained transformer reaches validation/test BLEU 0.4008/0.3572 and validation/test ChrF 0.6569/0.6380, while the stronger seq2seq reference run reaches 0.1284/0.1348 BLEU and 0.3491/0.3548 ChrF. On the shared validation/test evaluation, the transformer therefore leads the seq2seq model by 0.2724/0.2224 BLEU and 0.3078/0.2832 ChrF.

Model	Train	Val BLEU	Test BLEU	Val ChrF	Test ChrF
Opus-MT Transformer	100	0.4008	0.3572	0.6569	0.6380
Seq2Seq + Attn	2000	0.1284	0.1348	0.3491	0.3548

Table 7: Q4 capped Multi30k comparison between the pretrained transformer reference and the classical seq2seq baseline.

Figure 5 makes the same comparison easier to scan visually. The transformer leads the classical seq2seq model on both exported test metrics, and the gap is large enough that the model ranking is unambiguous even on the capped evaluation split.

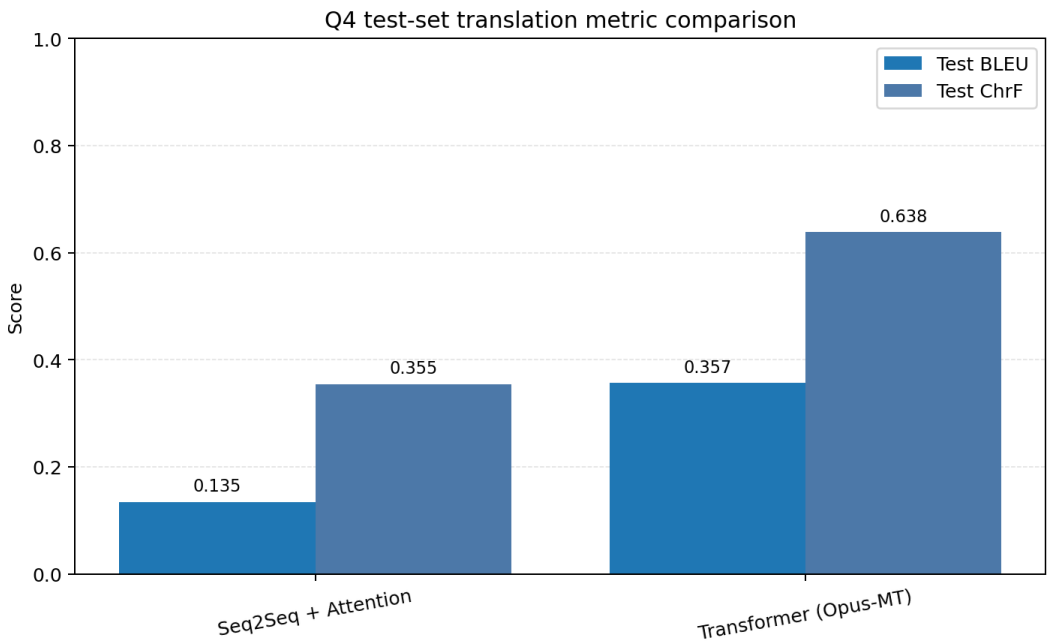

Figure 5: Q4 test-set BLEU and ChrF comparison for the stronger seq2seq baseline and the pretrained transformer reference.

The size of this gap still makes the main takeaway straightforward: under the current capped setup, the from-scratch seq2seq baseline improves materially with more training budget but is still not competitive with a strong pretrained translation checkpoint. At the same time, the comparison should not be oversold as a perfectly aligned training-budget experiment, because the transformer benefits from large-scale pretraining rather than the capped Multi30k train split.

5.4 Qualitative Analysis

The qualitative examples reinforce the metric gap. On test example `test_63`, the transformer exactly matches the reference translation for *Two men playing guitar in front of a large audience.*, which is the ideal behavior for a short caption translation task. Even its weaker exported example, `test_8`, remains understandable: it rewrites the running-man caption as a longer relative-clause construction that is less concise and less reference-like than the gold translation, but it still preserves most of the core meaning.

The seq2seq outputs show why the classical baseline still remains far behind despite the stronger run. Its best exported test example, `test_48`, captures much of the source meaning in *ein älterer mann spielt saxophon. ein videospiel.*, but it still introduces an incorrect content word and awkward sentence structure. Its weakest exported example, `test_94`, drifts badly into *drei chefköche, darunter auf dem rücken zu zu*, which is grammatical only in fragments and no longer faithful to the source sports caption. In other words, the seq2seq model now recovers more partial meaning than the weaker earlier run, but it still fails to sustain fluent, source-faithful translation consistently.

5.5 Discussion

The current Q4 evidence now supports a clearer conclusion than the earlier baseline-only draft. The pretrained transformer is the strongest finished translation system by a wide margin and should be treated as the main Q4 result, while the seq2seq model serves as the classical reference point that shows how difficult translation remains for small from-scratch models on limited data.

The remaining caveat is experimental alignment. Because the transformer is evaluated as a pretrained system rather than fine-tuned on Multi30k, a more budget-aligned next step would be either stronger seq2seq tuning or task-specific transformer fine-tuning on a shared train split. For the current report state, however, the direct comparison already answers the assignment question well enough to show that pretraining dominates the classical seq2seq baseline even after the stronger 2000-example seq2seq run is used.

6 Question 5: Language Modeling

6.1 Dataset Description

Question 5 uses a capped WikiText-2 setup drawn from the raw `wikitext-2-raw-v1` split [8]. The refreshed comparison evaluates 3,000 training lines, 400 validation lines, and 400 test lines, which correspond to 258,079, 32,527, and 35,059 tokens after preprocessing. This budget is still much smaller than a full language-model benchmark, so the reported perplexities should be read as a controlled comparison across a classical n-gram baseline, a compact recurrent neural baseline, and a practical pretrained transformer reference rather than as final large-budget results.

WikiText-2 remains appropriate because it contains longer-form Wikipedia prose rather than isolated short sentences. That makes it useful for testing both local collocations and longer-range continuation behavior. A later smaller LSTM rerun also exists elsewhere in the saved experiment outputs. This section focuses on the matched 3,000/400/400 held-out split so that the three finished Q5 models can be compared on the same validation and test data.

6.2 Model and Preprocessing

The data-trained Q5 baselines use the same lowercase word-level tokenization, the same minimum token-frequency threshold of 2, and the same unknown-token handling so that validation and test perplexity are measured against a consistent vocabulary. The classical baseline is a trigram language model with add-k smoothing and $\alpha = 0.1$, which provides a transparent count-based reference point for next-token prediction.

The second comparator is a compact two-layer word-level LSTM [4] with 128-dimensional embeddings and hidden states, dropout 0.2, tied input-output weights, and teacher-forced next-token training. It is still a modest recurrent model rather than a state-of-the-art language model. Even so, it is sufficient to test whether a learned hidden-state representation materially improves perplexity and generated-text fluency over the trigram baseline under the same capped dataset budget.

The third model is a practical GPT-style reference using pretrained `distilgpt2`, a compact member of the GPT-2 causal language-model family [10]. Unlike the trigram and LSTM models, this transformer is not trained from scratch on the capped Q5 split. Instead, it is evaluated directly on the same held-out WikiText-2 lines and prompted with the same seeded generation prefixes. That makes it a useful reference point for how much additional modeling strength a pretrained causal transformer contributes beyond the data-limited baselines.

6.3 Results

Table 8 reports the refreshed Q5 perplexity comparison. The trigram model reaches validation/test perplexity 1392.54/1388.72, the LSTM reduces those values to 286.06/279.46, and the pretrained distilGPT-2 reference further lowers them to 109.78/106.44.

Model	Validation Perplexity	Test Perplexity	Source Run
distilGPT-2	109.78	106.44	run_20260413_213856
LSTM	286.06	279.46	run_20260413_211945
Trigram add-k	1392.54	1388.72	run_20260413_202258

Table 8: Matched Q5 perplexity comparison across the finished trigram, LSTM, and GPT-style baselines on the capped WikiText-2 split.

Figure 6 shows the same ordering on a log-scaled y-axis so that the trigram, LSTM, and distilGPT-2 bars remain readable in a single chart. The pretrained transformer remains the strongest model on both validation and test, while the LSTM sits well below the classical trigram baseline but still well above distilGPT-2.

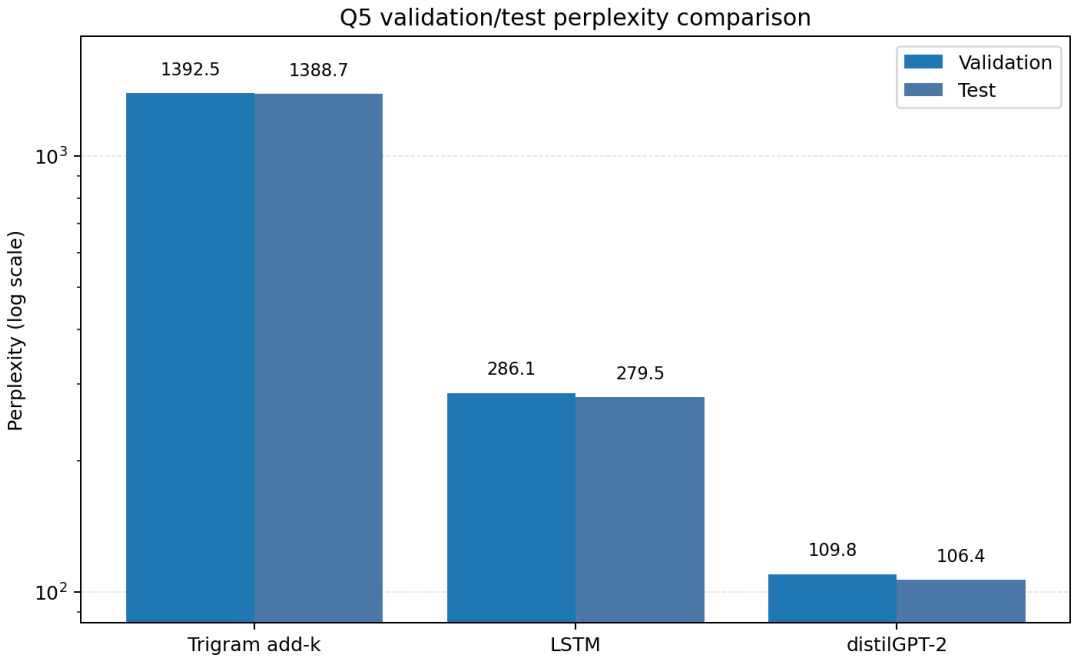

Figure 6: Q5 validation and test perplexity comparison on a log-scaled y-axis.

Two comparisons matter here. First, the LSTM cuts validation perplexity by 79.5% and test perplexity by 79.9% relative to the trigram baseline, which shows that even a compact recurrent language model is far better than sparse count-based context matching on held-out WikiText-2 continuations. Second, distilGPT-2 lowers validation perplexity by a further 61.6% and test perplexity by 61.9% relative to the matched LSTM result, making the pretrained transformer the strongest finished Q5 model.

6.4 Generated Text Samples

The exported generations are consistent with the perplexity comparison. The trigram model can maintain short-range lexical continuity for simple prompts, but it often stops early or drifts into topic fragments that are only locally plausible. Starting from the seed `the`, it generates the short fragment *the support of the season started .*, which is locally acceptable but not coherent in a broader discourse sense.

The LSTM continuations are still noisy, but they are longer and generally more structurally sustained under the same seed prompts. Its generated samples for the prompts "the," "in the," and "he was" continue for roughly 26–27 tokens, compared with a trigram average of 17.7 tokens across the same prompts. The LSTM still exhibits repetition and topic drift, yet it preserves local grammatical structure for longer spans than the sparse n-gram model.

The distilGPT-2 generations are slightly shorter on average than the LSTM samples (24.3 vs. 26.7 tokens), but they are noticeably smoother locally. For example, the prompt "he was" continues as *he was a little bit too late and it was time to try new things with my husband.*, which is still imperfect and not guaranteed to remain on-topic, yet it is substantially more fluent than the corresponding LSTM and trigram outputs.

6.5 Discussion

The finished Q5 comparison now makes the progression from classical to neural to pretrained-transformer modeling explicit. The trigram model remains valuable as a classical reference, but its high perplexity and unstable generations show that short fixed-context counts are not enough to model Wikipedia-style prose well. The LSTM then shows that even a compact recurrent model can reuse hidden-state context to reduce perplexity dramatically.

For the current report state, the trigram run should be kept as the classical baseline. The LSTM should be presented as the compact neural comparator, and distilGPT-2 should be presented as the strongest finished Q5 reference. If Q5 is reopened later, the most meaningful follow-up is a larger matched rerun or actual transformer fine-tuning rather than additional n-gram work.

7 Conclusion

This report now supports a clearer cross-task interim pattern: representation quality matters more than model family alone. When pretrained contextual or generative models are available, they are the strongest finished systems in every mature comparison or they at least define the strongest current baseline. At the same time, the results do not support a simple classical-versus-neural narrative, because lightweight classical baselines remain competitive and randomly initialized neural sequence models do not consistently outperform them.

For Question 1, DistilBERT is the best sentiment classifier on the matched IMDb comparison, but the gap over TF-IDF + SVM is modest relative to the much larger gap over the supervised-only BiLSTM. This indicates that the main gain comes from contextual pretraining rather than from using a neural architecture in the abstract. For Question 2, BERT again dominates the comparison and achieves the strongest overall and per-entity F1, while the feature CRF remains a strong non-transformer baseline and the BiLSTM-CRF underperforms both alternatives. For Question 3, the pretrained DistilBART summarizer outperforms TextRank on every exported lexical metric in the current capped comparison, suggesting that pretrained abstractive generation is more effective than purely extractive ranking for CNN/DailyMail-style summaries. For Question 4, the pretrained `Helsinki-NLP/opus-mt-en-de` system decisively outperforms the custom seq2seq-plus-attention baseline on the shared capped validation/test split, showing that the small from-scratch translation model is not competitive with a strong pretrained checkpoint under the current setup even though the training regimes are not perfectly matched.

Question 5 is now stronger than the earlier baseline-only and two-model states because the report includes a matched trigram-versus-LSTM-versus-GPT-style comparison. The trigram model remains informative as a classical reference, the LSTM reduces test perplexity from 1388.72 to 279.46, and the pretrained distilGPT-2 reference further lowers test perplexity to 106.44 while also producing smoother short continuations than the data-limited baselines. Across Questions 1 through 5, the most defensible project-level claim is therefore that pretrained models provide the strongest empirical results when direct comparisons exist, classical baselines remain essential reference points, and supervised neural models without strong pretraining or further tuning are not reliably superior by default.

The report is still provisional in a narrower sense. Question 4 now has a written comparison, but its train-budget mismatch means that a later fine-tuned transformer or stronger seq2seq rerun could sharpen the claim further, and Questions 3 and 5 may yet be extended with larger matched reruns or additional fine-tuning. Even so, the current written sections already support the central comparative lesson of the project: the main performance gains come from representation quality and pretraining, not from adopting a neural architecture by itself.

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